

Radio astronomy unit outline

Teacher planning guide

Guide students from their first horn telescope observations to a measured Galactic rotation curve. Compare their results with gravitational models.

Learning goal

Students measure hydrogen spectra and infer gas velocities at different distances from the Galactic center. They compare the rotation curve with a model dominated by a central mass. The mismatch motivates testing other mass distributions, including a dark matter halo.

How the investigation fits a semester

Introduce the telescope during the first week. Continue observations and analysis as students learn waves, circular motion, and gravitation. The investigation connects scientific models with measurements, uncertainty, and evidence.

When	Classroom focus
First week	Introduce the telescope, electromagnetic waves, hydrogen emission, coordinates, Stellarium, and Doppler shifts.
First month	Practice telescope operation. Observe Galactic quadrants II and III, then compare velocity measurements.
Before or during gravitation	Observe quadrant I. Use the tangent method to construct a rotation curve and evaluate mass models.

Preparation and assessment

Arrange access to a working horn telescope, a computer, and Stellarium. Plan observing dates when the required Galactic longitudes are visible.

Assess observation records, calculations, participation, and explanations. Finish with a class poster that compares measurements, uncertainties, and model predictions.

Adapted from the Astronomy in AP Physics 2018–19 unit plan. This guide describes teaching connections, not current exam alignment.

Build the foundations

Use these lessons during the first week. Adjust pacing to students' experience with graphs, waves, and coordinate systems.

Lesson 1 Horn Telescope Introduction

Observe the live spectrum and sketch the telescope system. Brainstorm how its components work.

Practice observation, graph reading, and scientific measurement. Review student work and participation.

Lesson 2 The Electromagnetic Spectrum

Read about waves and complete the exercises using $v = f\lambda$.

Connect frequency, wavelength, wave speed, and the electromagnetic spectrum. Assess the calculations.

Lesson 3 HI Hydrogen as a Radio Source

Read about neutral hydrogen, particle spin, photons, and energy transitions. Complete the photon-energy exercise.

Explain why hydrogen emits near 21 cm. Review the exercise and the student's explanation.

Lesson 4 Galactic Coordinates

Compare celestial and Galactic coordinates through reading, Stellarium, and exercises.

Use coordinates to describe telescope pointing. Assess the completed work.

Lesson 5 Using Stellarium

Explore how the sky changes with location and time. Find targets using celestial and Galactic coordinates.

Practice planning an observation. Assess students' records and coordinate choices.

Lesson 6 Data Analysis and the Doppler Shift

Read hydrogen spectra and identify frequency shifts. Calculate velocities from those shifts.

Connect spectral peaks to motion along the line of sight. Check signs, units, and uncertainty.

Collect data and test models

Lesson 7 Using the Horn Telescope

Practice operating the telescope before collecting the class survey. Record pointing, observing time, and the settings needed to interpret each spectrum.

Galaxy observation 1

Assign groups positions along the Galactic plane in quadrants II and III. Save spectra, measure peak frequencies, and calculate radial velocities.

Compare class results and look for systematic redshifts and blueshifts. Estimate how Earth's orbital motion could affect the measurements. Use circular motion and Doppler models to explain the pattern.

Assess each group's data, calculations, uncertainty estimates, and conclusion about Galactic rotation.

Galaxy observation 2

Assign groups positions in quadrant I. Collect spectra and identify the velocities needed for the tangent method. Use the geometry to relate these measurements to distances from the Galactic center.

Combine class measurements into a graph of orbital speed versus Galactocentric radius. Compare the measured curve with a Keplerian prediction. Discuss which assumptions fail and test alternative mass distributions.

Use gravitational and circular-motion models to interpret the results. Explain how additional unseen mass could account for the observations.

Final class poster

- State the question and describe the telescope measurements.
- Show the rotation curve with units, uncertainty, and a clear model comparison.
- Explain the tangent method and the assumptions used.
- Discuss limitations and whether the data support additional unseen mass.